# Package ‘confeR’

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**Title** confeR

**Description** Implements Bayesian conjugate analysis for federated inference as described in Degen et al. (2025) <doi:XXXXXXXXXXXX>.

**License** GPL-3

**Encoding** UTF-8

**Imports** invgamma, dplyr, tidyr, tibble

**Suggests** roxygen2

**NeedsCompilation** no

**RoxygenNote** 7.3.3

**URL** https://gitlab.uzh.ch/crsuzh/bayseq

**BugReports** https://gitlab.uzh.ch/crsuzh/bayseq/issues

**Depends** R (>= 3.5)

**LazyData** true

## R topics documented:

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`bayes_lin_reg_stats` *BCA linear regression summary stats*

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## Description

Takes in dataframe with (X, y) and returns sufficient statistics (X'X, X'y, y'y, n)

## Usage

```
bayes_lin_reg_stats(
  df,
  outcome,
  covariates,
  weights = NULL,
  k = 0,
  n_sites = 0
)
```

## Arguments

| | |
|---|---|
| `df` | Dataframe containing covariates X and outcome y. |
| `outcome` | Name of outcome y. |
| `covariates` | Vector of covariate names. |
| `weights` | Vector of weights for weighted linear regression. Default NULL. |
| `k` | Site number in case local intercepts are used. Default 0 corresponds to not using local intercepts. |
| `n_sites` | Number of sites. Default 0. |

## Value

List with transmitted summary statistics for each local site.

## Author(s)

Peter Degen

bca_iterate_sites *BCA iterate over local sites*

### Description

Pre-processing wrapper for testing BCA methods using individual participant data.

### Usage

```
bca_iterate_sites(
  model,
  family,
  data_split,
  use_local_intercepts,
  outcome = NULL,
  covariates = NULL,
  alpha = 0.05
)
```

### Arguments

| | |
|---|---|
| `model` | Formula object relating outcome to covariates. |
| `family` | Family to pass to glm() call, e.g. "gaussian". |
| `data_split` | List, each element containing a dataframe with the IPD from a local site. |
| `use_local_intercepts` | Logical. If true, use fixed site-specific intercepts for each local site. |
| `outcome` | Name of outcome y. |
| `covariates` | Vector of covariate names. |
| `alpha` | Numeric. Significance level for credible intervals. Default 0.05. |
| `center_name` | Character (optional). Name of covariate denoting site identity. Must be supplied if ‘use_local_intercepts‘ is TRUE. |

### Value

List with transmitted summary statistics for each local site.

### Author(s)

Peter Degen

bca_oneshot *BCA one-shot*

## Description

Perform one-shot computation of parameter estimates given transmitted summary statistics.

## Usage

```
bca_oneshot(
  sumstats,
  family,
  use_local_intercepts,
  use_local_variances = FALSE,
  epsilon = 1e-10,
  alpha = 0.05,
  covariates = NULL,
  glm_prior_lamda = 0
)
```

## Arguments

`sumstats` List of summary statistics from each local site, e.g. objbect returned by ‘bca_iterate_sites‘.

`family` Family to pass to glm() call, e.g. "gaussian".

`use_local_intercepts` Logical. If true, use fixed site-specific intercepts for each local site.

`use_local_variances` Logical. If true, use fixed site-specific residual variances for each local site. Default FALSE.

`epsilon` Numeric. Regularization for prior hyperparameters. Default: 1e-10.

`alpha` Numeric. Significance level for credible intervals. Default: 0.05.

`glm_prior_lamda` Numeric (optional). Prior for diagonal precision matrix in normal model, known variance case. See also Equation 7 in Jonker, Pazira and Coolen (2024). Default: 0.

`center_name` Character (optional). Name of covariate denoting site identity. Must be supplied if ‘family‘ is ‘gaussian‘.

## Value

List of oneshot parameter estimates

## Author(s)

Peter Degen

forest_box *Multivariate forest plot wrapper*

## Description

Wrapper that computes pbox and makes forest plot in one call.

## Usage

```
forest_box(
  params_oneshot,
  sumstats,
  family,
  use_local_intercepts,
  pboxes = NULL,
  center_name = "site",
  remove_intercept = FALSE,
  alpha = 0.05,
  remove_sigma2 = TRUE,
  ...
)
```

## Arguments

| | |
|---|---|
| `params_oneshot` | Parameters returned by ‘bca_oneshot‘. |
| `sumstats` | List of summary statistics from each local site, e.g. obbject returned by ‘bca_iterate_sites‘. |
| `family` | Family to pass to glm() call, e.g. "gaussian". |
| `use_local_intercepts` | Logical. If true, use fixed site-specific intercepts for each local site. |
| `center_name` | Character. Name of covariate denoting site identity. |
| `alpha` | Numeric. Significance level for credible intervals. Default 0.05. |
| `...` | Further arguments passed to ‘prepare_forest_plot‘. |

## Value

ggplot2 object.

## Author(s)

Peter Degen

`forest_plot` *Multivariate forest plot*

## Description

Creates a forest plot with a facet for each covariate. Optionally prints Box's prior predictive tail probability for each site.

## Usage

```
forest_plot(
  df_forest,
  pboxes = NULL,
  outfile = NULL,
  alpha = 0.05,
  nrow = 1,
  inline_plot = FALSE,
  use_log_scale = TRUE,
  order_box = TRUE
)
```

## Arguments

| | |
|---|---|
| `df_forest` | Dataframe returned by 'prepare_forest_plot'. |
| `pboxes` | Optional numeric vector of Box p-values, contained in list returned by 'box_check_all_sites'. |
| `outfile` | Optional character. If not null, save figure in this file. |
| `alpha` | Numeric. Significance level for credible intervals. Default 0.05. |
| `nrow` | Numeric. (default: 1) |
| `inline_plot` | Logical. If TRUE, adjust width and height of inline plot in Jupyter notebooks using 'options()'. (default: FALSE) |
| `use_log_scale` | Logical. If TRUE, print pbox as an S-value, i.e. -log2(pbox) (default: FALSE) |
| `order_box` | FALSE Logical. If TRUE, order sites by p-Box. (default: FALSE) |

## Value

ggplot2 object.

## Author(s)

Peter Degen

`get_pred_prob` *Box prior predictive tail probability*

## Description

Check if estimate from specific site is compatible with posterior from all other centers.

## Usage

```
get_pred_prob(
  center_identity,
  sumstats,
  n_sites,
  reduced_params,
  use_local_intercepts,
  remove_intercept = FALSE
)
```

## Arguments

- `center_identity` The id of the site to be checked
- `sumstats` List of summary statistics from each local site, e.g. obbject returned by ‘bca_iterate_sites‘.
- `n_sites` Number of sites.
- `reduced_params` Reduced posterior parameters returnd by ‘get_reduced_params‘.
- `use_local_intercepts` Logical. If true, use fixed site-specific intercepts for each local site.
- `remove_intercept` Logical. If true, ignore intercept in pbox calculation. Default: FALSE.
- `covariates` Vector of covariate names.

## Value

The Box prior predictive tail probability.

## Author(s)

Peter Degen

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`get_reduced_params` *Get reduced params*

---

## Description

Use Reverse-Bayes to get reduced parameters obtained by leaving out one local site from the full posterior.

## Usage

```
get_reduced_params(center_identity, params_oneshot, sumstats, family)
```

## Arguments

| | |
|---|---|
| `center_identity` | The id of the site to be removed. |
| `params_oneshot` | Parameters returned by bca_oneshot. |
| `sumstats` | List of summary statistics from each local site, e.g. obbject returned by ‘bca_iterate_sites‘. |
| `family` | Family to pass to glm() call, e.g. "gaussian". |

## Value

List with reduced parameter estimates.

## Author(s)

Peter Degen

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`nurses_hom` *Nurses hom data*

---

## Description

This simulated data set consists of 1000 data points representing nurses from 25 hospitals. The outcome of interest is job-related stress among nurses (standardized) and the covariates are age (standardized), experience (standardized), sex (0 = male, 1 = female), the type of ward in which the nurse works (0 = general care, 1 = special care), and hospital (1, 2, ..., 25).

## Usage

```
nurses_hom
```

### Format

## ‘nurses_hom‘

**data_ipd** Individual participant data. A data frame with 1000 rows and 6 variables:

- **gender** factor: 0 = male, 1 = female
- **age** dbl: Standardized age
- **experience** dbl: Standardized experience
- **wardtype** factor: 0 = general care, 1 = special care
- **stress** dbl: Standardized stress levels
- **hospital** factor: Hospital ID, 1 to 25

**summary_stats_hom** List of summary statistics for each hospital: (X’X, X’y, y’y, n). Use this for models with a global intercept.

**summary_stats_het** List of summary statistics for each hospital: (X’X, X’y, y’y, n). Use this for models with a site-specific intercepts.

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`pred_check` *Box prior predictive tail probabilities for all sites*

---

### Description

Calculate p_Box for all sites.

### Usage

```
pred_check(
  params_oneshot,
  sumstats,
  family,
  use_local_intercepts,
  center_name = NULL,
  remove_intercept = FALSE
)
```

### Arguments

`params_oneshot` Parameters returned by ‘bca_oneshot‘.

`sumstats` List of summary statistics from each local site, e.g. obbject returned by ‘bca_iterate_sites‘.

`family` Family to pass to glm() call, e.g. "gaussian".

`use_local_intercepts` Logical. If true, use fixed site-specific intercepts for each local site.

`center_name` Character. Name of covariate denoting site identity.

`remove_intercept` Logical. If true, ignore intercept in pbox calculation. Default: FALSE.

`data_split` The data split by local sites.

`covariates` Vector of covariate names.

`n_sites` Number of sites.

### Value

List with pboxes and reduced parameters for all sites.

### Author(s)

Peter Degen

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## `prepare_forest_plot` *Prepare forest plot*

---

### Description

Get clean dataframe with local glm parameters from summary statistics for all sites plus the federated BCA estimates.

### Usage

```
prepare_forest_plot(df_bca, sumstats, family, alpha = 0.05)
```

### Arguments

| | |
|---|---|
| `df_bca` | Dataframe with BCA estimates returned by ‘tidy_results‘. |
| `sumstats` | List of summary statistics from each local site, e.g. obbject returned by ‘bca_iterate_sites‘. |
| `alpha` | Numeric. Significance level for credible intervals. Default 0.05. |

### Value

Dataframe with parameter estimates and confidence intervals for all sites and covariates.

### Author(s)

Peter Degen

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## `tidy_results` *BCA tidy results*

---

### Description

Tidy results from one-shot

### Usage

```
tidy_results(params_oneshot, use_local_intercepts)
```

## Arguments

params_oneshot Parameters returned by ‘bca_oneshot‘.

use_local_intercepts
Logical. If true, use fixed site-specific intercepts for each local site.

## Value

Dataframe

## Author(s)

Peter Degen

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`transform_X` *transform X*

---

## Description

Take data vector X and convert factors with numeric. Binary factors will be encoded by 0, 1. Factors with more than two levels will be transformed using one-hot encoding, dropping one level to avoud multicollinearity.

## Usage

```
transform_X(X)
```

## Arguments

X Design matrix X with covariates

## Value

Transformed design matrix X

## Author(s)

Peter Degen

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